



Saint Columban College
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Packets Primer

Packets Primer 



Medium

Forensics

picoCTF 2022

pcap

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Description

Download the packet capture file and use packet analysis software to find the flag.

- [Download packet capture](#)

Hints 

1

30,321 users solved



88% Liked



picoCTF{FLAG}

Submit Flag

Author: The Analyst: Hyposelenia

Challenge: Packets Primer

Category: Forensics

Date: 03/31/26



I. Objective

The objective of this challenge is to understand how network data is captured and analyzed using **packet analysis tools**.

The goal is to inspect a packet capture file and **extract hidden information (the flag)** from network traffic.

II. Background

When computers communicate over a network (like the internet), they don't send data all at once.

Instead, they break it into small pieces called **packets**.

These packets travel across networks and are reassembled at the destination.

In cybersecurity and forensics, analyzing these packets can reveal:

- messages
- credentials
- hidden data

This challenge uses a **.pcap file**, which stores captured network traffic.

III. Tool Used

- **Kali Linux (Oracle VirtualBox)** - Used as the working environment
- **Wireshark** - Used to analyze packet capture files

IV. Methodology

1. Launched **Kali Linux**



2. Downloaded the file using:

wget <link_address>

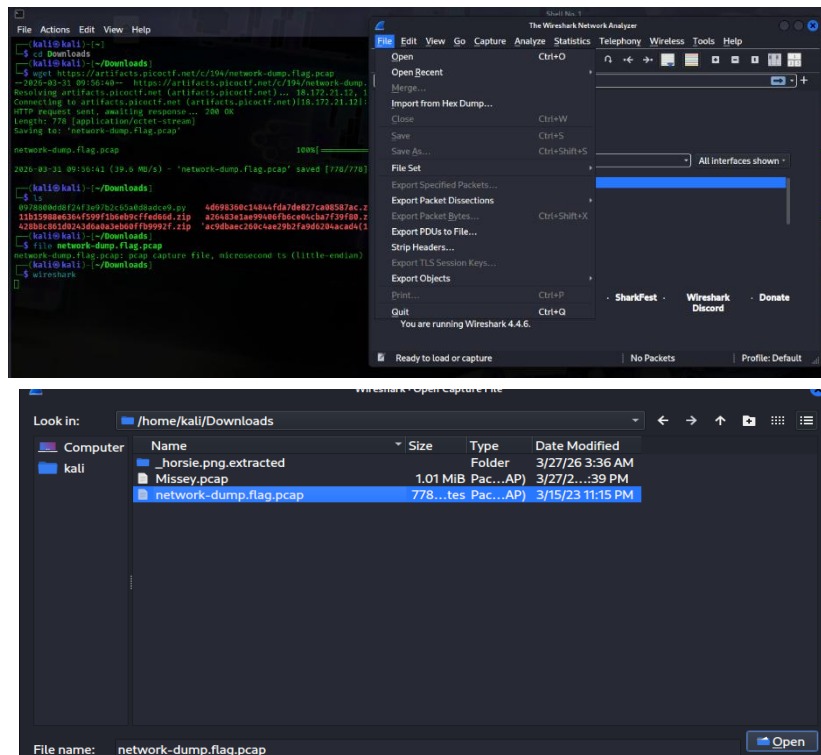
```
(kali㉿kali)-[~]
$ cd Downloads
(kali㉿kali)-[~/Downloads]
$ wget https://artifacts.picoctf.net/c/194/network-dump.flag.pcap
--2026-03-31 09:56:40-- https://artifacts.picoctf.net/c/194/network-dump.flag.pcap
Resolving artifacts.picoctf.net (artifacts.picoctf.net) ... 18.172.21.12
Connecting to artifacts.picoctf.net (artifacts.picoctf.net)|18.172.21.12|:443
HTTP request sent, awaiting response... 200 OK
Length: 778 [application/octet-stream]
Saving to: 'network-dump.flag.pcap'

network-dump.flag.pcap                               100%[=====]
2026-03-31 09:56:41 (39.6 MB/s) - 'network-dump.flag.pcap' saved [778/778]
```

3. Observed that the file had a .pcap extension (packet capture file)

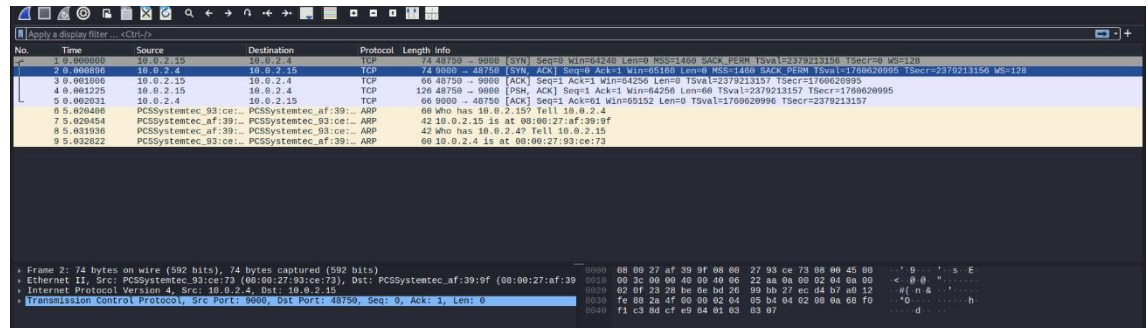
```
(kali㉿kali)-[~/Downloads]
$ file network-dump.flag.pcap
network-dump.flag.pcap: pcap capture file, microsecond ts (little-endian) - version 2.4.0
```

4. Opened the file using **Wireshark**

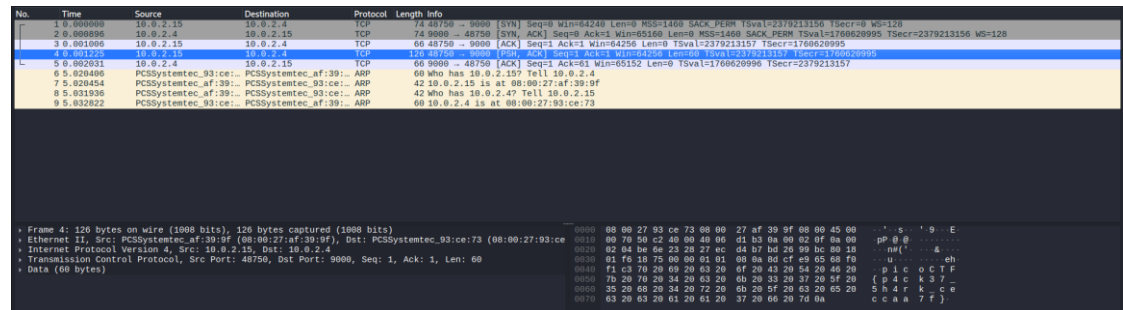




5. Browsed through the packet list and examined IP sources and traffic



6. Identified a suspicious source containing readable data

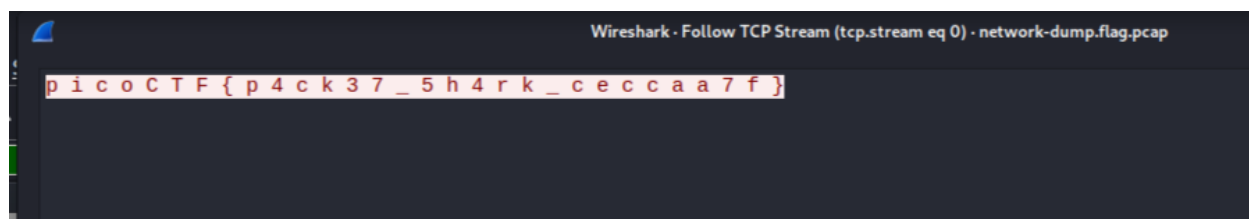


7. Right-clicked the packet

8. Selected:

Follow → TCP Stream

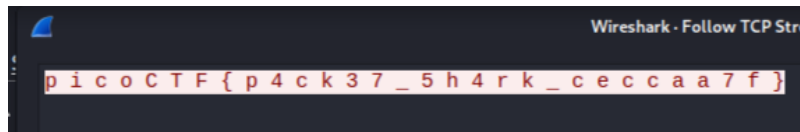
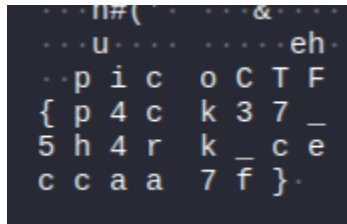
9. Viewed the reconstructed communication



10. Located and copied the flag

V. Result

picoCTF{p4ck37_5h4rk_ceccaa7f}



VI. Explanation

What is a Packet?

A packet is:

A small piece of data sent between computers

Real-life analogy:

Imagine sending a long message

Instead of sending it all at once:

- You break it into **small pieces**
- Send them one by one

Each piece = **packet**

How do computers communicate?

Computers talk using:

- networks (internet)
- protocols (rules for communication)

Think of it like:

- People talking = conversation
- Computers talking = packets

What is Wireshark?

Wireshark is a tool that:



Lets you see and analyze packets

What is “Follow Stream”?

When packets are sent, they are:

- broken into pieces
- scattered across many packets

Follow Stream” means:

“Put all the pieces back together into one message”

I believe in real-world cybersecurity:

- Analysts inspect packet captures
- They look for:
 - passwords
 - suspicious activity
 - data leaks

VII. Conclusion

This challenge demonstrates how network communication can be analyzed to uncover hidden information.

By examining packet data and reconstructing communication streams, the flag was successfully retrieved.

This reinforces a key concept in digital forensics:

Network traffic can reveal valuable information when properly analyzed.

— **The Analyst: Hyposelenia**